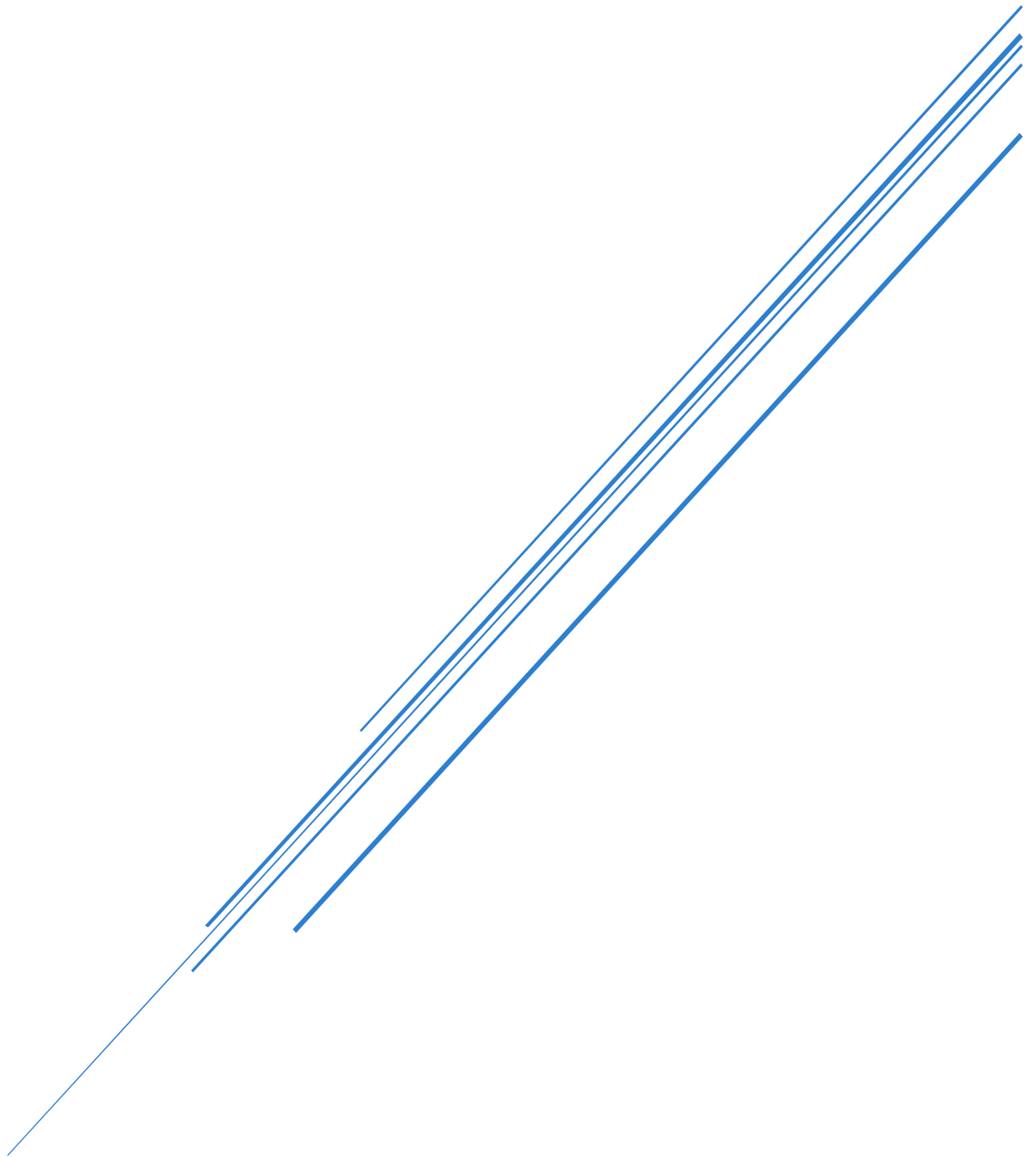


PROJECT PROPOSAL

Digital Transformation Case Study



github.com/ImEasoon

Table of Contents

1.0 Problem Statement.....	3
1.1 Inefficient Physical Filing System.....	3
1.2 Limited Data Accessibility and Operational Visibility.....	3
1.3 Data Integrity & Accuracy Risk.....	3
2.0 Proposed Solution	4
2.1 Digital Data Collection.....	4
2.1.1 Key Objectives	4
2.1.2 Benefits.....	4
2.2 Centralised Database	4
2.2.1 Key Objectives	4
2.2.2 Benefits.....	5
2.3 Real-time Production Dashboard	5
2.3.1 Key Objectives	5
2.3.2 Benefits.....	5
2.4 Reporting & Analysis Platform	5
2.4.1 Key Objectives	5
2.4.2 Benefits.....	6
3.0 System Architecture	7
3.1 System Requirements	7
3.1.1 Functional Requirements	7
3.1.2 Non-Functional Requirements.....	8
3.2 Recommended Tools & Technologies.....	9
3.3 System Design	10
3.3 System Workflow	11
3.4 System Interface	12
4.0 Implementation.....	14
4.1 Project Planning.....	14
4.2 Required Resources	14
4.2.1 Manpower	14
4.2.2 Hardware	14
4.2.3 Software	14

4.3 Budget Estimation	15
4.3.1 Cost Calculation.....	15
4.4 Priority of Limited Resources.....	16
4.5 Practical & Realistic Implementation Approach	16
5.0 Risks & Challenges	17
6.0 Future Improvements.....	19
6.1 Automated Data Collection.....	19
6.2 Predictive Analytics	19
6.3 Enterprise Resource Planning (ERP) Integration	19
6.4 Artificial Intelligence (AI) Agent Deployment	20
7.0 Conclusion	21

1.0 Problem Statement

1.1 Inefficient Physical Filing System

The production information currently is stored in physical and highly dependent on manual paperwork. This brings a chain of operational inefficiencies, including:

- The historical production data is hard to retrieved and managed.
- The slow preparation of reports and meetings by management.

Result: increased administrative workload and delays in decision-making.

1.2 Limited Data Accessibility and Operational Visibility

The data has not been organised and converted into a usable format in a timely manner. This will limit the ability to monitor operational performance and management control.

- The lack of real-time visibility of production status.
- The data is limited for management analysis.
- The preparation of the report and the planning for the meeting have been delayed.

Result: communication delays and misunderstandings caused.

1.3 Data Integrity & Accuracy Risk

The manual processes increase the risk of inconsistencies and errors in production records.

- The risk of missing records and inaccurate documentation.
- The lack of a standardised data validation procedure.

Result: reduced data reliability in reports and analyses.

2.0 Proposed Solution

To address the issues identified, the Company XXX needs to digitise production data and implement a manufacturing monitoring system. It aims to replace paperwork with a practical digital workflow.

The system should consist of the proposed solution below:

2.1 Digital Data Collection

Operators may replace manual paperwork with electronic data entry. Relevant information can be entered directly into the proposed system via electronic devices such as computers, tablets and smartphones.

2.1.1 Key Objectives

- Reduce the dependence on manual paperwork.
- Enhanced the efficiency of data entry.
- Provide faster access to production information.
- Minimise missing or inaccurate records.

2.1.2 Benefits

- Standardised data collection procedures.
- Improved data accuracy.
- Reduced human mistake and negligence.

2.2 Centralised Database

All production-related data including quantity, machine status, downtime records and so on will be securely stored on a database server and accessible only to authorised users.

2.2.1 Key Objectives

- Centralise the management of production records.
- Improve the historical data search functionality.

- Ensure data consistency and security.

2.2.2 Benefits

- Easier access to historical production data.
- More efficient data management.
- Reduced the risk of record disappear.
- Improved information sharing across departments.

2.3 Real-time Production Dashboard

A real-time production dashboard uses visualisation such as charts, indicator and summaries to provide operators with the ability to monitor manufacturing operations including progress, performance, downtime and more.

2.3.1 Key Objectives

- Enable real-time production status monitoring.
- Improve operational transparency.

2.3.2 Benefits

- Stay informed about the latest production status.
- Quickly identify production issues.
- Optimised production control.
- Improved operational efficiency.

2.4 Reporting & Analysis Platform

A reporting and analysis platform will automatically generate production reports and analytical insights based on the collected data.

2.4.1 Key Objectives

- Enhance the efficiency of report and meeting preparation.
- Provide intuitive data summaries.
- Support management analysis.

2.4.2 Benefits

- Optimised data analysis process.
- Reasonable production planning.
- Clearly discover the operational trends and issues.
- Strengthen decision-making according to reports.

3.0 System Architecture

3.1 System Requirements

3.1.1 Functional Requirements

The manufacturing monitoring system should achieve the following functions.

1. User Authentication and Access Control

- The system shall allow users to log in using a username and password.
- The system shall restrict access by unauthorized people.

2. Digital Data Collection

- The system shall support data entry using smart devices.
- The system shall only allow relevant personnel to enter production data.
- The system shall validate the input data before submission.

3. Centralised Database Management

- The system shall store all the production data in a centralised database.
- The system shall only allow authorised users to retrieve historical production data.
- The system shall support backup and recovery functions.

4. Real-Time Production Monitoring

- The system shall display the real-time production status on the dashboard.
- The system shall refresh and update data in real time.

5. Notification and Alert Management

- The system shall timely inform specific users when abnormal situations happen.

6. Reporting and Analysis

- The system shall automatically generate summaries and graphical reports based on collected data.
- The system shall allow users to export and download reports in specific formats.

- The system shall support search and filtering function.

7. Activity Logging

- The system shall log user activities.

3.1.2 Non-Functional Requirements

The manufacturing monitoring system should perform as expected condition below.

1. Performance

- The system shall display the real-time data updates within a few seconds.
- The system shall generate reports within an acceptable response time.
- The system shall support simultaneous access by multiple users without compromising performance.

2. Reliability

- The system shall continue to operate with minimal system downtime.
- The system shall provide automatic data backup and recovery mechanisms.
- The system shall prevent data loss during system failures.

3. Security

- The system shall only grant access to authenticated users.
- The system shall encrypt sensitive data during transmission.
- The system shall restrict access to specific groups and roles.

4. Usability

- The system interface shall be user-friendly and easy to learn.
- The system shall support responsive design to adapt to different screen sizes of smart devices.

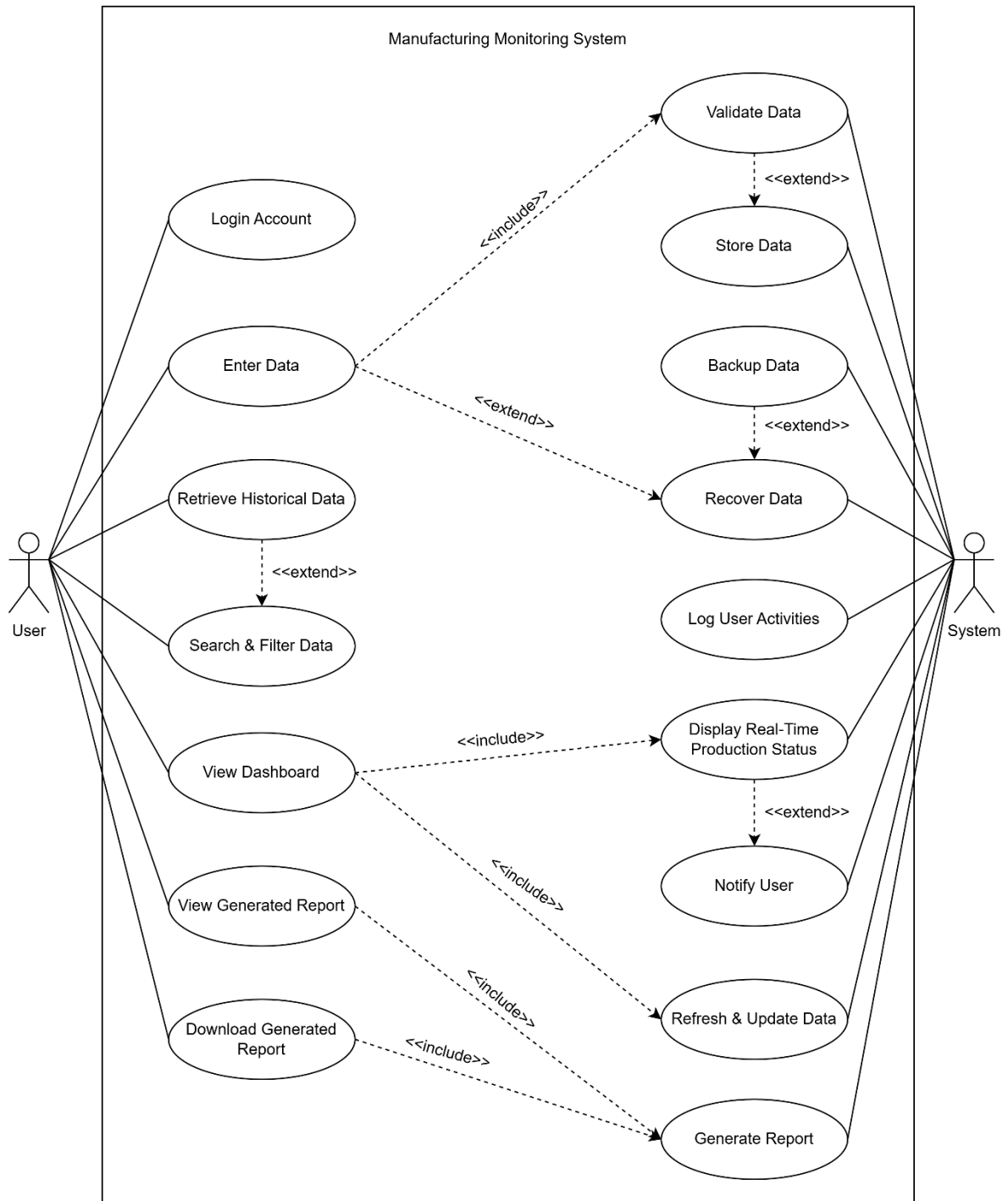
5. Compatibility

- The system shall be compatible with modern web browsers.
- The system shall be executable on electronic devices.

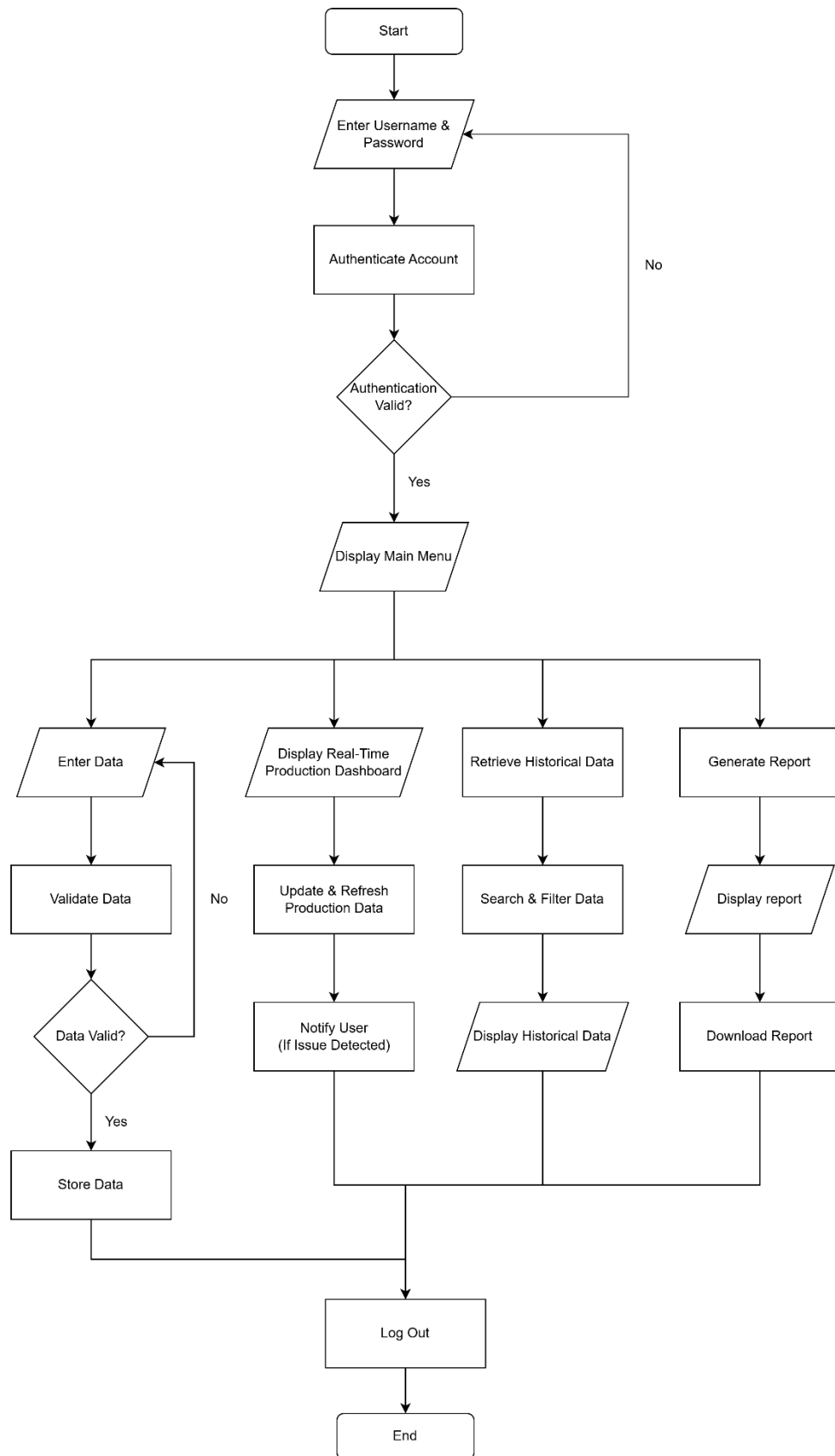
3.2 Recommended Tools & Technologies

Architecture Layer	Recommendation
Data Entry	Web Application Form
Frontend Framework	React
Backend API	Node.js
Database	MySQL
Hosting	Cloud Service
Data Visualisation	Power BI

3.3 System Design



3.3 System Workflow



3.4 System Interface



Logo

Manufacturing Monitoring System

  Log Out

DASHBOARD

DATA ENTRY

REPORTS & ANALYSIS

Production Information

Submit

Date*

18/05/2026



Shift*

Morning



Machine ID*

M-101



Operator Name*

Admin

Product Code*

P-101

Quantity*

100

Rejected Quantity*

0

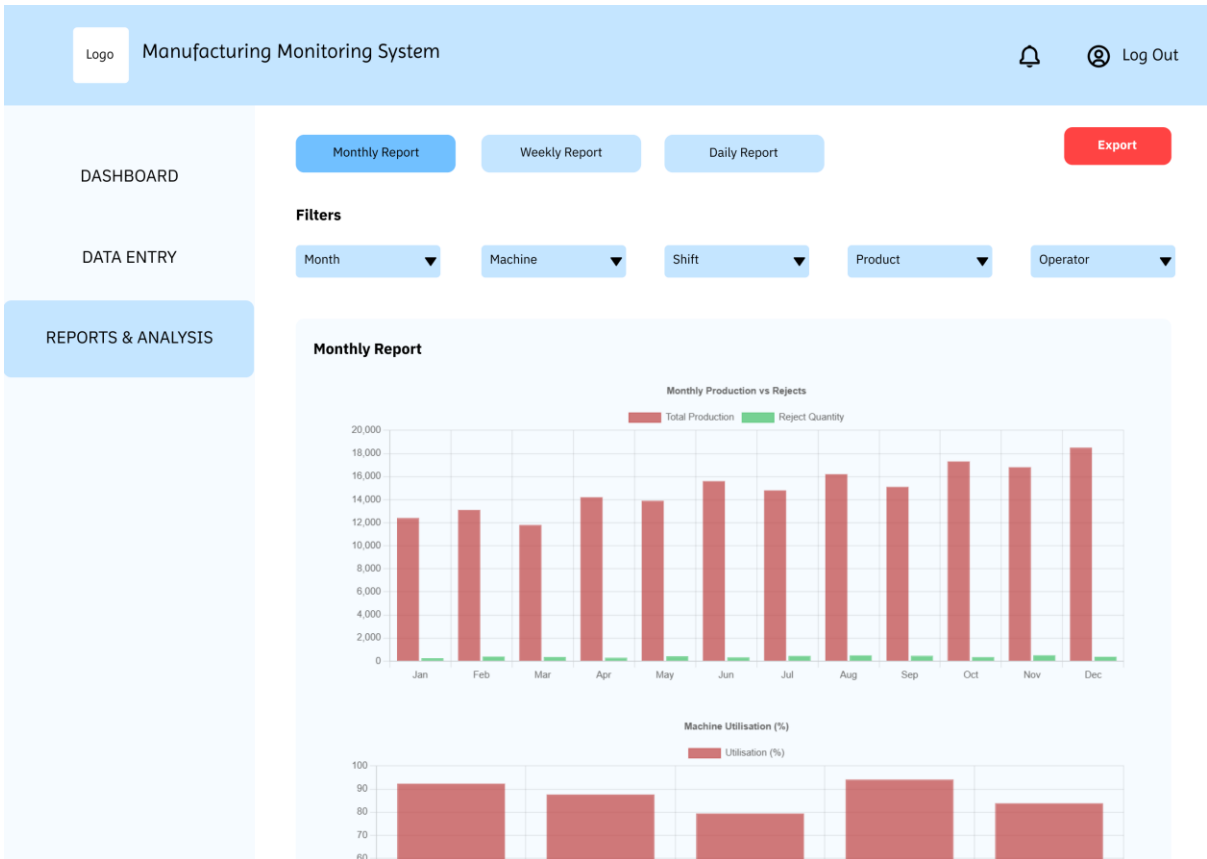
Machine Status*

Running



Downtime Reason

None



4.0 Implementation

4.1 Project Planning

[Manufacturing Monitoring System] GANTT CHART												
Task	Week(s)											
	1	2	3	4	5	6	7	8	9	10	11	12
Requirement Review & Finalisation												
System Design Finalisation												
Development												
Testing												
Pilot Run												
Deployment												

4.2 Required Resources

4.2.1 Manpower

Role	Responsibility	Project Involvement
Project Manager	Coordination	25%
Business Analyst	Requirement Gathering	25%
System Developer	System Development	100%
UI / UX Designer	User Interface Design	30%
Database Engineer	Database Establishment and Setup	40%
Testing & QA	System Testing and Feedback	20%
IT Support	Deployment and Maintenance	10%

4.2.2 Hardware

Item	Quantity
Tables / Desktop PC (inc. keyboard, mouse, barcode scanner)	Per production line
Cloud Hosting	1
Network Infrastructure	WLAN

4.2.3 Software

Item	Purpose
Database	Data Storage

Dashboard Software	Data Visualisation and Reporting
Frontend Framework	User Interface
Backend Server	APIs Connection

4.3 Budget Estimation

Item	Estimated Cost
Development	RM 22,815
Hardware	RM 10,300
Hosting	RM 680
Dashboard License	RM 600
Training	RM 550
Total	RM 34,945

4.3.1 Cost Calculation

- Development Cost = RM 22,815
 - Project Manager = $\text{RM } 3,500 * 25\% * 3 \text{ Months} = \text{RM } 2,625$
 - Business Analyst = $\text{RM } 2,200 * 25\% * 3 \text{ Months} = \text{RM } 1,650$
 - System Developer = $\text{RM } 3,500 * 100\% * 3 \text{ Months} = \text{RM } 10,500$
 - UI/UX Designer = $\text{RM } 2,500 * 30\% * 3 \text{ Month} = \text{RM } 2,250$
 - Database Engineer = $\text{RM } 3,000 * 40\% * 3 \text{ Month} = \text{RM } 3,600$
 - Testing & QA = $\text{RM } 2,500 * 20\% * 3 \text{ Month} = \text{RM } 1,500$
 - IT Support = $\text{RM } 2,300 * 10\% * 3 \text{ Month} = \text{RM } 690$

- Hardware Cost = RM 10,300
 - Tablet = $\text{RM } 1,000 * 2 \text{ Qty} = \text{RM } 2,000$
 - Desktop PC = $\text{RM } 2,500 * 2 \text{ Qty} = \text{RM } 5,000$
 - Monitor = $\text{RM } 1,200 * 2 \text{ Qty} = \text{RM } 2,400$
 - Keyboard & Mouse = $\text{RM } 100 * 2 \text{ Qty} = \text{RM } 200$
 - Barcode Scanner = $\text{RM } 50 * 4 \text{ Qty} = \text{RM } 200$
 - Router = $\text{RM } 250 * 2 \text{ Qty} = \text{RM } 500$

- Hosting Cost = RM 680

- Azure Database Hosting = Approximate RM 600
 - D2ds v4 = $\text{RM } 500 * 1 \text{ Months} = \text{RM } 500$
 - 200 GB = $\text{RM } 100 * 1 \text{ Months} = \text{RM } 100$
- Backup Storage = Approximate RM 80
 - 200 GB = $\text{RM } 80 * 1 \text{ Month} = \text{RM } 80$
- Dashboard License Cost = RM 600
 - Power BI Pro = $\text{RM } 60 * 10 \text{ Users} = \text{RM } 600$
- Training Cost = Approximate RM 550
 - Trainer Fee = $\text{RM } 500 * 1 \text{ Week} = \text{RM } 500$
 - User Manual = RM 50

4.4 Priority of Limited Resources

As a hypothesis, the system should still include the following features as the highest priority although resources for this project are limited.

- Digital Production Data Entry
- Centralised Database
- Basic Real-Time Production Dashboard

These functions eliminate most dependencies on paperwork and improve visibility for management to analyse data and make decisions.

4.5 Practical & Realistic Implementation Approach

To avoid big failures in the production environment, system deployment should start on a small scale.

- Phrase 1: The system should be piloted only on the smallest production lines with the fewest operators.
- Phrase 2: After a period of using, the system should be improved according to the feedback collected.
- Phrase 3: The system should be gradually rolled out to all production lines.
- Phrase 4: Operating procedures should be standardised to make the system easier for users to use.

5.0 Risks & Challenges

During the implementation of the proposed system, Company XXX might face and need encounter the following challenges.

1. Acceptance Issues

- Some operators may be unfamiliar with the new proposed system and worry the change will increase their workloads.

2. Poor Network

- Since the system relies on network, the internet instability may cause interruptions in real-time data updates and affect production monitoring.

3. Incorrect Data Entry

- Operators may occasionally enter incorrect values, input incomplete or duplicate information which can compromise data accuracy and affect subsequent analysis.

4. Budget Limitation

- The system requires a significant investment in software development, hardware purchase, staff training and maintenance and limited budget may delay the implementation.

5. Device Compatibility

- Older production equipment may not be compatible with modern technologies and system integration.

6. Cybersecurity Risks

- The system stores the business data digitally will increase the risk of unauthorised access, malware attack and data leakage.

Risk & Challenge	Solution
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Acceptance Issues	Staff Training and Demonstration.
Poor Network	Local Caching
Incorrect Data Entry	Input Validation Rules
Budget Limitation	Prioritise Critical Parts
Device Compatibility	Upgrade Equipment Gradually
Cybersecurity Risks	Encryption, Firewall Establishment and Strong Access Control

6.0 Future Improvements

After the digital transformation of the production process is successfully implemented, the proposed system can be further expanded and integrated with modern technologies.

6.1 Automated Data Collection

The system can connect to IoT sensors and image recognition models to reduce the need of manual data entry.

- IoT sensors can directly collect data and automatically populate the web application forms.
- Image recognition models can help convert information from original documents into digital format. This digital data provides management with clearer and more comprehensive statistical information.

6.2 Predictive Analytics

The system can be enhanced with predictive analytics capabilities which identify patterns and trends to forecast production performance, equipment downtime and material consumption.

- Forecasting models can estimate future production output and forecast requirements to optimise inventory management. It also predicts equipment maintenance schedules and detects anomalies at an early stage to reduce unexpected shutdown.

6.3 Enterprise Resource Planning (ERP) Integration

The system can be integrated with an ERP system to synchronise business workflow and automatically update information across other departments.

- ERP system integration can automatically update inventory using production output data to ensure accurate stock levels as well as trigger purchase request when the raw material stock runs low.

6.4 Artificial Intelligence (AI) Agent Deployment

The system can deploy AI agents to automate daily tasks and receive intelligence assistance using natural language.

- AI agents can reduce workload by automatically generating reports, summarizing collected data, assisting system users via chatbot and providing decision support and insights.

7.0 Conclusion

In a nutshell, the proposed digital transformation should focus on the statement below:

1. Reduce manual paperwork.
2. Centralise production data.
3. Improve the visibility of real-time production data.
4. Support reporting and analysis.

The system implementation should start small and gradually move toward smart manufacturing capabilities once the system has been fully refined.